(c) Bigram Modeling

$$p(w_i / w_{i-1}) = \frac{count(w_{i-1}, w_i)}{count(w_{i-1})}$$

<s> I am Sam </s>
<s> Sam I am </s>
<s> I do not like green eggs and hams </s>

$$p(I / \langle s \rangle) = \frac{count(\langle s \rangle, I)}{count\ \langle s \rangle} = \frac{2}{3}$$

$$p(am / I) = \frac{count(I, am)}{count\ I} = \frac{2}{3}$$

$$p(sam / am) = \frac{count(am, sam)}{count(am)} = \frac{1}{2}$$

$$p(\langle /s \rangle / Sam) \quad \frac{count(sam, \langle /s \rangle)}{count(sam)} = \frac{1}{2}$$

$$p(Sam / \langle s \rangle) = \frac{1}{3}$$

$$p(hams / and) = \frac{1}{1}$$

$$p(I / sam) \quad \frac{1}{2}$$

$$p(\langle /s \rangle / hams) = \frac{1}{1}$$

$$p(\langle /s \rangle / am) \quad \frac{1}{2}$$

$$p(do / I) = \frac{1}{3}$$

$$p(not / do) = \frac{1}{1} = 1$$

$$p(like / not) = \frac{1}{1} = 1$$

$$p(green / like) = \frac{1}{1} = 1$$

$$p(eggs / green) = \frac{1}{1} = 1$$

$$p(and / eggs) = \frac{1}{1}$$

Test Sentence: I am Sam.

$$p(am|I) = \frac{2}{3}$$

$$p(sam|am) = \frac{1}{2}.$$

$$p(\text{I am Sam}) = \frac{2}{3} * \frac{1}{2} = \frac{1}{3}. = 0.33$$

$\boxed{33\%}$

Laplace Smoothing

$$p(w_i|w_{i-1}) = \frac{Count(w_{i-1}, w_i) + 1}{Count(w_i - 1) + |V|}$$

$$|V| = 12$$

$$p(I/\langle s \rangle) = \frac{Count(\langle s \rangle, I) + 1}{Count \langle s \rangle + |V|}$$

$$= \frac{2+1}{3+12} = \frac{3}{15} = 0.2$$

$\boxed{20\%}$

Trigram Modelling

$$p(w_i|w_{i-1}, w_{i-2}) = \frac{Count(w_{i-2}, w_{i-1}, w_i)}{Count(w_{i-2}, w_{i-1})}$$

$$p(am|I, \langle s \rangle) = \frac{count(\langle s \rangle, I, am)}{Count(\langle s \rangle, I)}$$

$$= \frac{1}{2}$$